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Proximity detection device

Abstract

A proximity detection device with a detection distance for detecting an object separated to some extent from an operation surface is provided. The proximity detection device includes a proximity detection unit that has a piezoelectric body and first and second electrodes disposed in contact with the piezoelectric body to detect proximity of an object, a signal applying unit that causes the proximity detection unit to perform capacitance detection and ultrasonic transmission and/or ultrasonic reception by applying a plurality of signals of different frequencies to at least one of the first and second electrodes, and a charge measurement unit connected to at least one of the first and second electrodes to measure electric charge.

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Background/Summary

CLAIM OF PRIORITY (1) This application is a Continuation of International Application No. PCT/JP2022/009264 filed on Mar. 3, 2022, which claims benefit of Japanese Patent Application No. 2021-077787 filed on Apr. 30, 2021. The entire contents of each application noted above are hereby incorporated by reference.

BACKGROUND OF THE INVENTION
1. Field of the Invention
(1) The present invention relates to a proximity detection device.
2. Description of the Related Art
(2) In general, the following touch panels have been used. That is, a touch panel has a pressure detection function and includes a sensor unit with a capacitive sensor and a piezoelectric sensor that is stacked on a back side of the capacitive sensor or that partially shares a component layer with the capacitive sensor, a capacitance detection circuit connected to the capacitive sensor of the sensor unit to detect a touch and a touch position in accordance with a capacitance change and to send an electric signal to a host, a charge amplifier connected to the piezoelectric sensor of the sensor unit to convert a charge signal into a voltage signal, a sample-and-hold circuit that is connected to the charge amplifier and that transmits an output of the charge amplifier while the capacitance detection circuit deactivates the capacitive sensor whereas holds and transmits, when the deactivation of the capacitive sensor is changed to activation of the capacitive sensor by the capacitance detection circuit, an output of the charge amplifier obtained immediately before the change, and an AD converter that is connected to the sample-and-hold circuit and that digitally converts an output of the sample-and-hold circuit and sends the output to the host (refer to, Japanese Unexamined Patent Application Publication No. 2019-194791).
(3) Here, a touch panel with a pressure detection function may detect a touch and a touch position on a touch surface, but the touch panel may not detect an object, such as a hand, that is some distance away from the touch surface. In other words, a distance at which an object may be detected in a direction away from an operation surface, such as the touch surface, is limited.
SUMMARY OF THE INVENTION
(4) The present invention provides a proximity detection device with a detection distance for detecting an object separated to some extend from an operation surface.
(5) The proximity detection device according to an embodiment of the present invention includes a proximity detection unit that has a piezoelectric body and first and second electrodes disposed in contact with the piezoelectric body to detect proximity of an object, a signal applying unit that causes the proximity detection unit to perform capacitance detection and ultrasonic transmission and/or ultrasonic reception by applying a plurality of signals of different frequencies to at least one of the first and second electrodes, and a charge measurement unit connected to at least one of the first and second electrodes to measure electric charge.

(6) A proximity detection device with a detection distance for detecting an object separated to some extend from an operation surface may be provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a diagram illustrating an example of a proximity detection device according to an embodiment;
(2) FIG. **2** is a diagram illustrating a configuration in a cross-section viewed from a II-II arrow in FIG. **1**;
(3) FIG. **3** is a flowchart of an example of a process executed by an MPU;
(4) FIG. **4**A is a diagram illustrating a modification of an intersection;
(5) FIG. **4**B is a diagram illustrating a modification of the intersection;
(6) FIG. **4**C is a diagram illustrating a modification of the intersection;
(7) FIG. **4**D is a diagram illustrating a modification of the intersection;
(8) FIG. **5**A is a diagram illustrating a modification of the intersection;
(9) FIG. **5**B is a diagram illustrating a modification of the intersection;
(10) FIG. **5**C is a diagram illustrating a modification of the intersection;
(11) FIG. **5**D is a diagram illustrating a modification of the intersection;
(12) FIG. **6**A is a diagram illustrating a modification of the intersection;
(13) FIG. **6**B is a diagram illustrating a modification of the intersection;
(14) FIG. **7**A is a diagram illustrating a modification of the intersection;
(15) FIG. **7**B is a diagram illustrating a modification of the intersection;
(16) FIG. **7**C is a diagram illustrating a modification of the intersection; and
(17) FIG. **8** is a diagram illustrating an example of a proximity detection device according to a modification of the embodiment.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

(18) The following is a description of an embodiment to which a proximity detection device according to the present invention is applied.

Embodiment

(19) FIG. **1** is a diagram illustrating an example of a proximity detection device **100** according to an embodiment. FIG. **2** is a diagram illustrating a configuration in a cross-section viewed from a II-II arrow in FIG. **1**. In the following, a description will be made while the XYZ coordinate system is defined. A direction parallel to an X axis (X direction), a direction parallel to a Y axis (Y direction), and a direction parallel to a Z axis (Z direction) are orthogonal to one another. For convenience of explanation, a side in an −Z direction may be referred to as a lower side and a side in a +Z direction as an upper side, but this does not represent a universal vertical relationship. Furthermore, a plan view means to view an XY plane. Moreover, lengths, thicknesses, etc. of each component may be exaggerated in the following to make a configuration easier to understand.

(20) The proximity detection device **100** includes a proximity detection unit **110**, an MUX (multiplexer) **120**, a transmitter circuit **130**, a receiver circuit **140**, a timer **145**, and an MPU (Micro Processing Unit) **150**. The MUX **120**, the transmitter circuit **130**, and the MPU **150** are an example of a signal applying unit. The signal applying unit causes the proximity detection unit **110** to perform capacitance detection and ultrasonic transmission and/or ultrasonic reception by applying signals of different frequencies to at least first electrodes **111** or second electrodes **112**.

(21) The proximity detection device **100** has an operation surface **100**A. The operation surface **100**A serves as a reference plane when the proximity detection device **100** detects the proximity of an object, and is a surface of a panel, such as a housing of an electronic device including the proximity detection device **100**, for example. The electronic device may be any electronic device

that includes a touch panel, for example, a smartphone or a tablet computer. The proximity detection unit **110** is disposed on a back side of the operation surface **100**A, and the operation surface **100**A is located on a front side of the proximity detection unit **110**. Here, a case where an object is a hand of a user of the electronic device including the proximity detection device **100** will be described.
(22) The proximity detection device **100** detects the proximity of a user's hand to the operation surface **100**A. Here, proximity means approach of a hand without touching the operation surface **100**A, or a touch on the operation surface **100**A with a hand.
(23) The proximity detection device **100** detects an amount of charge corresponding to capacitance between a hand and the proximity detection unit **110**, and performs profiling and image detection based on the amount of charge to calculate a position of the hand that is an object. The calculation of a hand position based on capacitance means that a position of a hand in contact (touch) with the operation surface **100**A and a position of the hand when the hand is not in contact with the operation surface **100**A but very close to the operation surface **100**A may be calculated. The hand position calculated based on capacitance may be represented in three dimensions.
(24) Furthermore, the proximity detection device **100** transmits ultrasonic waves toward the hand from a plurality of portions of the proximity detection unit **110** and calculates distances from the operation surface **100**A to a plurality of points on the hand based on round-trip times until reflected waves are received. In other words, the proximity detection unit **110** performs an ultrasonic transmission followed by an ultrasonic reception by reflected waves. The proximity detection device **100** may further calculate distribution of hand positions using the calculated distances.
(25) There is a limit to a range in which changes in capacitance due to hand proximity may be detected in a direction away from the operation surface **100**A. The calculation of a hand position based on the round-trip times of ultrasonic waves is difficult when the hand position is too close to the operation surface **100**A. Therefore, the proximity detection device **100** calculates a distance to the hand using ultrasonic waves when the distance from the operation surface **100**A to the hand is longer than a predetermined distance, and calculates the distance to the hand using capacitance when the distance from the operation surface **100**A to the hand is equal to or shorter than the predetermined distance. In determining whether the distance from the operation surface **100**A to the hand is longer than the predetermined distance, as an example, a plurality of distances from the operation surface **100**A to a plurality of points on the hand are obtained based on round-trip times of ultrasonic waves, and it is determined whether an average of the plurality of distances is longer than the predetermined distance. The predetermined distance is from 3 cm to 10 cm as an example. Furthermore, instead of an average of a plurality of distances, it may be determined whether a minimum of a plurality of distances is longer than the predetermined distance or whether a distance at a certain point is longer than the predetermined distance.
(26) The proximity detection device **100** may calculate a two-dimensional profile representing a two-dimensional distribution of a hand shape or a three-dimensional image representing a three-dimensional distribution of the hand shape when the user's hand approaches the operation surface **100**A. Such a two-dimensional profile or a three-dimensional image of the hand shape may be obtained based on distances from the operation surface **100**A to a plurality of points on the hand, which may be calculated based on, among other things, round-trip times which are from transmissions of ultrasonic waves from a plurality of portions of the proximity detection unit **110** to receptions of reflected waves.
(27) The two-dimensional profile of the hand shape represents, for example, a two-dimensional distribution of a hand position obtained according to distribution of distances in the Z direction from the operation surface **100**A to the plurality of positions of the hand in an XZ plane in a certain Y coordinate or in a YZ plane in a certain X coordinate. Furthermore, the three-dimensional image of the hand shape represents a three-dimensional distribution of a hand position obtained according to distribution of distances in the Z direction from the operation surface **100**A to the plurality of

positions of the hand.
(28) Moreover, the proximity detection device **100** shares the same detection unit that detects a distance to the hand using ultrasonic waves and that detects a hand position using capacitance, and uses the MUX **120**, the transmitter circuit **130**, the receiver circuit **140**, the timer **145**, and the MPU **150** as the common detection unit.
(29) <Proximity Detection Unit **110**>
(30) The proximity detection unit **110** has piezoelectric bodies and first and second electrodes arranged in contact with the piezoelectric bodies to detect proximity of an object. It is assumed here that the term "piezoelectric body" in this application refers to a substance having piezoelectricity, and examples of the piezoelectric body include an electret having piezoelectricity. In this embodiment, the proximity detection unit **110** includes first electrodes **111**, second electrodes **112**, piezoelectric bodies **113**, and substrates **114** and detects the proximity of a hand as an object. The first electrodes **111** are linear electrodes (electrode lines) extending in the X direction, and the plurality of first electrodes **111** are arranged at equal intervals in the Y direction. The X direction is an example of a first direction and the Y direction is an example of a second direction. The second electrodes **112** are linear electrodes (electrode lines) extending in the Y direction, and the plurality of second electrodes **112** are arranged at equal intervals in the X direction. The first and second electrodes **111** and **112** are spaced in the Z direction and intersect in plan view, and the piezoelectric bodies **113** are arranged between the first and second electrodes **111** and **112** at the intersections **110**A which are portions where the first and second electrodes **111** and **112** intersect in plan view. In other words, the first and second electrodes **111** and **112** are arranged to be in contact with the piezoelectric bodies **113**, and the first and second electrodes **111** and **112** sandwich the piezoelectric bodies **113** therebetween.
(31) The first and second electrodes **111** and **112** are used both to detect a distance to a hand using ultrasonic waves and to detect a position of the hand using capacitance. The piezoelectric bodies **113** are used to detect a distance to a hand using ultrasonic waves. In other words, the proximity detection unit **110** also has a function of ultrasonic detection since the electrodes for capacitance detection also serve as electrodes for the piezoelectric bodies.
(32) The intersections **110**A where the first electrodes **111** and the second electrodes **112** intersect in plan view are arranged in a matrix as illustrated in FIG. **1**. Since the substrates **114** are disposed on the first electrodes **111** and under the second electrodes **112**, a cross-sectional configuration of each of the intersections **110**A includes the substrate **114**, a corresponding one of the second electrodes **112**, a corresponding one of the piezoelectric bodies **113**, a corresponding one of the first electrodes **111**, and the substrate **114** which are stacked from bottom to top, as illustrated in FIG. **2**.
(33) The first and second electrodes **111** and **112** are wire electrodes made of metal, such as copper or aluminum, as an example. As an example, the proximity detection unit **110** may be fabricated by preparing the substrate **114** with the plurality of first electrodes **111** on one surface and the substrate **114** with the plurality of second electrodes **112** on one surface, and pasting the two substrates **114** together with the piezoelectric bodies **113** sandwiched therebetween at the intersections **110**A. At the intersections **110**A, the piezoelectric bodies **113** are arranged between the first and second electrodes **111** and **112**, but outside the intersections **110**A, the first and second electrodes **111** and **112** are insulated by an insulating layer or the like.
(34) The piezoelectric bodies **113** are disposed to generate ultrasonic vibrations. Ultrasonic waves are used because it is easy to measure a distance to a hand located directly above the piezoelectric bodies **113** by radiating highly directional ultrasonic waves directly above the individual piezoelectric bodies **113** (in the +Z direction). As the piezoelectric bodies **113**, elements which produce distortion by application of a voltage, such as piezoelectric elements, may be used, for example. At the intersections **110**A, the first and second electrodes **111** and **112** are disposed on and under the piezoelectric body **113**, respectively, and therefore, by applying an AC signal for ultrasonic waves between the first and second electrodes **111** and **112**, the piezoelectric bodies **113**

may resonate and transmit ultrasonic waves directly above the piezoelectric bodies **113**.
(35) The AC signal for ultrasonic waves is an example of a second frequency signal for ultrasonic transmission and at a frequency that may resonate the piezoelectric bodies **113** arranged between the first and second electrodes **111** and **112**. The frequency of the AC signal for ultrasonic waves is several tens of kHz to several hundreds of kHz as an example, and the piezoelectric bodies **113** vibrate at a frequency equal to the frequency of the AC signal. By applying an AC signal for ultrasonic waves between the first and second electrodes **111** and **112**, the piezoelectric bodies **113** may resonate to generate ultrasonic waves at a desired frequency.
(36) Instead of the piezoelectric bodies **113**, electrets with piezoelectric properties may be used. In this case, the electrets may generate ultrasonic waves in the same way as the piezoelectric bodies **113** by applying an AC signal for ultrasonic waves while being sandwiched between the first and second electrodes **111** and **112**.
(37) The substrates **114** may be flexible or rigid type wiring substrates, insulating sheets, or the like. The proximity detection unit **110** may be made transparent for visible light. In this case, the first and second electrodes **111** and **112** are made of a transparent conductive material, such as ITO (Indium Tin Oxide), the piezoelectric bodies **113** are transparent, and the two substrates **114** are transparent.
(38) <MUX **120**>
(39) The MUX **120** is connected to the first and second electrodes **111** and **112** via wiring, as well as to the transmitter circuit **130** and the receiver circuit **140**. The MUX **120** selects one or more of the first electrodes **111** and one or more of the second electrodes **112** to switch the first and second electrodes **111** and **112** to be connected to the transmitter circuit **130** and the receiver circuit **140** in a time-series manner. The switching of the selection of the first and second electrodes **111** and **112** by the MUX **120** is performed under control of the MPU **150**.
(40) <Transmitter Circuit **130**>
(41) When detecting a distance to a hand by ultrasonic waves, the transmitter circuit **130** outputs, under control of the MPU **150**, an AC signal for ultrasonic waves between the individual first electrodes **111** and the individual second electrodes **112** via the MUX **120** and also outputs the AC signal to the timer **145**. When performing position detection using capacitance, the transmitter circuit **130** outputs, under control of the MPU **150**, an AC signal of a frequency for capacitance detection to the individual first electrodes **111** or the individual second electrodes **112** via the MUX **120**. Since the detection of a distance to a hand using ultrasonic waves and the position detection using capacitance are separately performed, for example, in a time-division manner, the transmitter circuit **130** selectively applies an AC signal for ultrasonic waves and an AC signal of a frequency for capacitance detection to the individual first electrodes **111** or the individual second electrodes **112**.
(42) The AC signal of the frequency for capacitance detection is an example of a first frequency signal for capacitance detection, and the AC signal for ultrasonic waves is an example of a second frequency signal for ultrasonic wave transmission. The frequency of the AC signal of a frequency for capacitance detection is several tens of kHz to several hundreds of kHz similarly to the AC signal for ultrasonic waves as an example as long as the frequency of the AC signal of a frequency for capacitance detection is out of the resonance frequency of the piezoelectric bodies **113**. This is because resonance of the piezoelectric bodies **113** is suppressed when the position detection is performed based on capacitance.
(43) <Receiver Circuit **140**>
(44) A charge measurement unit is connected to at least the first electrodes **111** or the second electrodes **112** so as to measure an electric charge. In this embodiment, the receiver circuit **140** corresponds to the charge measurement unit. When a distance to a hand is detected using ultrasonic waves, the receiver circuit **140** acquires, under control of the MPU **150**, waveforms generated by electric charges between the individual first electrodes **111** and the individual second electrodes **112**

via the MUX **120** and outputs the waveforms to the timer **145**. When the position detection is performed using capacitance, the receiver circuit **140** detects, under control of the MPU **150**, an amount of charge corresponding to the capacitance between the individual first electrode **111** and the individual second electrode **112** via the MUX **120** and outputs the amount of charge to the MPU **150**.

(45) <Timer **145**>

(46) When a distance to a hand is detected by ultrasonic waves, the timer **145** measures, under control of the MPU **150**, time differences between waveforms of an AC signal supplied from the transmitter circuit **130** and waveforms supplied from the receiver circuit **140** as round-trip times of the ultrasonic waves for the individual first electrodes **111** and the individual second electrodes **112**. The timer **145** outputs the round-trip times measured for the individual first electrodes **111** and the individual second electrodes **112** to the MPU **150**.

(47) <MPU **150**>

(48) The MPU **150** has a main controller **151**, a calculation unit **152**, and a memory **153**. The MPU **150** is realized by a computer that includes a CPU (Central Processing Unit), a RAM (Random Access Memory), a ROM (Read Only Memory), an I/O interface, and an internal bus. The main controller **151** and the calculation unit **152** are functional blocks indicating functions of programs executed by the MPU **150**. Furthermore, the memory **153** functionally indicates a memory of the MPU **150**.

(49) The main controller **151** is a processing unit that controls processing of the MPU **150** and executes processes other than those executed by the calculation unit **152**, for example.

(50) The calculation unit **152** calculates a distance between an object (e.g., hand) and the proximity detection unit **110**. The MPU **150** causes the MUX **120** to switch a selection of the first and second electrodes **111** and **112** when detecting the distance to the hand using ultrasonic waves or when detecting a position using capacitance. The MPU **150** is connected to the first and second electrodes **111** and **112** selected by the MUX **120**, and therefore, the calculation unit **152** is connected to the first and second electrodes **111** and **112** selected by the MUX **120** via the transmitter circuit **130** or the receiver circuit **140**.

(51) The signal applying unit selectively applies a first frequency signal for capacitance detection and a second frequency signal for ultrasonic transmission. The calculation unit **152** obtains a result of capacitance detection and/or ultrasonic detection based on the signal selected by the signal applying unit and the charge measured by the receiver circuit **140** (charge measurement unit), and in addition, calculates a distance between an object and the proximity detection unit **110** based on the obtained result. Specifically, when the first frequency signal is selected by the signal applying unit, the calculation unit **152** obtains the result of the capacitance detection based on the amount of charge measured by the receiver circuit **140** and calculates a distance between the object and the proximity detection unit **110** based on the obtained result. On the other hand, when the second frequency signal is selected by the signal applying unit, the calculation unit **152** calculates the distance between the object and the proximity detection unit **110** based on a period of time from ultrasonic transmission to ultrasonic reception based on the charge measured by the receiver circuit **140**. In other words, the calculation unit **152** is a common calculation unit that is capable of determining results of both the capacitance detection and the ultrasonic detection using the amount of charge of the first electrodes **111** and/or the second electrodes **112** measured by the charge measurement unit (receiver circuit **140**).

(52) Furthermore, the signal applying unit selects the second frequency signal and causes the proximity detection unit **110** to transmit and/or receive ultrasonic waves, and selects the first frequency signal and causes the proximity detection unit **110** to perform capacitance detection when the distance between the object and the proximity detection unit **110** calculated by the calculation unit **152** is a predetermined distance or shorter. Specifically, the MPU **150**, as an example, controls the transmitter circuit **130** to output an AC signal for ultrasonic waves between

the individual first electrodes **111** and the individual second electrodes **112** while switching a selection of the first electrodes **111** and the second electrodes **112** performed by the MUX **120** in a time-series manner, and obtains round-trip times from the timer **145**. The calculation unit **152** then calculates distances from the operation surface **100**A to portions of a hand directly above the intersections **110**A.

(53) When an average of all the distances is longer than a predetermined distance, for example, the signal applying unit causes the proximity detection unit **110** to transmit and receive ultrasonic waves so that a distance to the hand is calculated using the ultrasonic waves. Furthermore, when the calculation unit **152** determines that the average of all the distances is equal to or shorter than the predetermined distance, the signal applying unit causes the proximity detection unit **110** to perform capacitance detection so that a position of the object is calculated using capacitance. Note that it is not necessarily the case that the determination is made using the average of all the distances, and the determination may be made based on a minimum of all the distances or a determination as to whether a distance between the object and the proximity detection unit **110** at a certain point is shorter than a predetermined distance may be made. Furthermore, in this embodiment, the timer **145** is provided separately from the MPU **150**, but the MPU **150** itself may have a function of measuring time, and in this case, the timer **145** is not required.

(54) When detecting the distance to the hand using ultrasonic waves, the MPU **150** controls the transmitter circuit **130** to output an AC signal for ultrasonic waves between the individual first electrodes **111** and the individual second electrodes **112** while the MUX **120** switches a selection of the first and second electrodes in a time-series manner, causes the receiver circuit **140** to acquire a waveform produced by the electric charge, and obtains round-trip times from the timer **145**.

(55) The calculation unit **152** calculates distances from the operation surface **100**A to points where the ultrasonic waves are reflected based on the round-trip times and a sound speed. The points where the ultrasonic waves are reflected are portions of the hand located directly above the piezoelectric bodies **113** at the intersections **110**A of the first and second electrodes **111** and **112** selected by the MUX **120**.

(56) Furthermore, based on the obtained distances, the calculation unit **152** may further detect a two-dimensional profile of a hand shape or a three-dimensional image of the hand shape. Specifically, the calculation unit **152** detects a two-dimensional profile or a three-dimensional image of the object, that is, the hand, based on the amount of charge measured by the receiver circuit **140** (charge measurement unit) and/or the period of time between ultrasonic transmission and ultrasonic reception. By detecting the two-dimensional profile or the three-dimensional image, the shape of the hand may be recognized and a movement of the user's hand may be detected.

(57) When performing the position detection using capacitance, the MPU **150** controls the transmitter circuit **130** to output an AC signal of a frequency for capacitance detection between the individual first electrodes **111** and the individual second electrodes **112** while the MUX **120** switches a selection of the first electrodes **111** and the second electrodes **112** in a time-series manner, and also controls the receiver circuit **140** to detect capacitance obtained from charge between the individual first electrodes **111** and the individual second electrodes **112**.

(58) Note that the change of a detection method using the predetermined distance is merely an example, and the MPU **150** may perform switching between a detection of the distance to the hand by ultrasonic waves and a position detection using capacitance in a time-division manner. In this case, the signal applying unit performs switching between the first and second frequency signals in a time-division manner. By executing the two types of detection method in a time-division manner, a detection of the distance to the hand using ultrasonic waves and a detection of the hand position using capacitance may always be performed, regardless of the distance from the operation surface **100**A to the hand.

(59) The memory **153** stores programs and data required by the main controller **151** and the calculation unit **152** to perform the processing described above, the round-trip times input to the

MPU **150** from the timer **145**, the distances calculated by the calculation unit **152**, the capacitance, and data representing the two-dimensional profile or the three-dimensional image of the hand shape.
(60) <Processing Executed by MPU **150**>
(61) FIG. **3** is a flowchart of an example of a process executed by the MPU **150**.
(62) When the process is started, the calculation unit **152** calculates a distance to a hand using ultrasonic waves (step S**1**). The calculation unit **152** calculates distances from the operation surface **100**A to portions of the hand directly above the individual intersections **110**A. The process in step S**1** is performed to determine whether to detect a distance to the hand using ultrasonic waves or a position using capacitance.
(63) The MPU **150** determines whether an average of all the distances is longer than a predetermined distance (step S**2**).
(64) When determining that the average of all the distances is longer than the predetermined distance (S**2**: YES), the MPU **150** detects the distance to the hand using ultrasonic waves (step S**3**). This is because a hand position is too far away to be detected using capacitance, and therefore, the calculation is based on round-trip times of the ultrasonic waves. Note that the details of the detection of the distance to the hand using ultrasonic waves have been described above and are omitted here.
(65) The calculation unit **152** detects a two-dimensional profile or a three-dimensional image of a hand shape based on the distance obtained in step S**2** (step S**4**). By this, a two-dimensional profile or a three-dimensional image of the hand approaching the operation surface **100**A is obtained.
(66) After the process in step S**4**, the MPU **150** determines whether to terminate a series of processes (step S**5**). It is determined that the series of processes is to be terminated in step S**5** when a power source of an electronic device including the proximity detection device **100** is turned off.
(67) When the MPU **150** determines that the series of processes is not to be terminated (S**5**: NO), the flow returns to step S**1**. This is in order to continue the process according to a subsequent position of the hand.
(68) Furthermore, when determining in step S**2** that the average of all the distances is not longer than the predetermined distance (S**2**: NO), the MPU **150** performs the position detection using capacitance (step S**6**). This is because a position of the hand is too close to be determined using round-trip times of ultrasonic waves, and therefore, the position is determined using capacitance. Note that, when the MPU **150** terminates the process in step S**6**, the flow proceeds to step S**5**.
(69) As described above, the proximity detection device **100** has the piezoelectric bodies **113** at the intersections **110**A of the first and second electrodes **111** and **112**, and when the position of the hand is equal to or shorter than the predetermined distance, position detection is performed using the capacitance obtained from the charge of the first and second electrodes **111** and **112**, and when the hand position is longer than the predetermined distance, the distance to the hand is calculated based on the round-trip times of the ultrasonic waves transmitted by driving the piezoelectric bodies **113**. When the detection of the distance to the hand using ultrasonic waves is employed, a distance much farther than is possible with the capacitance-based position detection may be detected.
(70) Therefore, the proximity detection device **100** with a detection distance that may detect an object that is some distance away from the operation surface **100**A may be provided.
(71) Furthermore, the proximity detection device **100** shares the same detection unit that detects the distance to the hand using ultrasonic waves and that detects the hand position using electrostatic capacitance. The common detection unit includes the MUX **120**, the transmitter circuit **130**, the receiver circuit **140**, the timer **145**, and the MPU **150**. In particular, the receiver circuit **140** may serve as the charge measurement unit, and therefore, the charge detection commonly required for the capacitance detection and the ultrasonic wave detection may be performed by the single component. Therefore, the detection of the distance to the hand using ultrasonic waves and the

detection of the hand position using capacitance may be executed by the same circuit, and a hand position near the operation surface **100**A based on capacitance and a hand position at some distance away from the operation surface **100**A using ultrasonic waves may be obtained with a simple configuration. In addition, since the same detection unit detects the distance to the hand using ultrasonic waves and detects the hand position using capacitance, the two different detection methods may attain the same detection accuracy. This is a solution to a problem that a device configuration is complicated and large when the distance to the hand is detected using ultrasonic waves and the hand position is detected using capacitance by different detection units.

(72) Furthermore, the proximity detection unit **110** is configured such that the first and second electrodes **111** and **112** sandwich the piezoelectric bodies **113** therebetween, and therefore, an AC signal may be easily applied to the piezoelectric bodies **113** by using the first and second electrodes **111** and **112** for capacitance detection. Note that the proximity detection unit **110** may be configured such that the first and second electrodes **111** and **112** sandwich piezoelectric electrets therebetween. Furthermore, since the MUX **120** connected to the first and second electrodes **111** and **112**, the transmitter circuit **130**, the receiver circuit **140**, and the MPU **150** are used for capacitance detection, the proximity detection device **100** also capable of detecting ultrasonic waves may be realized by only adding the timer **145** and changing the program executed by the MPU **150**. This is a solution to the problem that the device configuration is complicated and becomes large when the detection of the distance to the hand using ultrasonic waves and the detection of the hand position using capacitance are performed by different proximity detection units.

(73) Furthermore, since the proximity detection unit **110** receives ultrasonic waves that are reflected waves after transmitting ultrasonic waves, a proximity detection unit for transmission and a proximity detection unit for reception are not required to be separately provided, and the single proximity detection unit **110** may perform transmission and reception of ultrasonic waves and the transmission and the reception of ultrasonic waves may be realized with a simple configuration.

(74) Furthermore, the signal applying unit selectively applies the first frequency signal which is for capacitance detection and the second frequency signal which is for ultrasonic transmission. In this embodiment, the MPU **150**, the MUX **120**, and the transmitter circuit **130**, which correspond to the signal applying unit, selectively apply an AC signal for capacitance detection and an AC signal for transmission of ultrasonic waves. The calculation unit **152** obtains a result of capacitance detection and/or ultrasonic detection based on the signal selected by the signal applying unit (MPU **150**, MUX **120**, and transmitter circuit **130**) and the measured charge, and in addition, calculates a distance between the object, that is, the hand, and the proximity detection unit **110** based on the obtained result. Therefore, a simple configuration can be achieved in which one MUX **120**, one transmitter circuit **130**, and one receiver circuit **140** may be used for both the capacitance detection and the ultrasonic detection.

(75) In addition, since the signal applying unit (MPU **150**, MUX **120**, and transmitter circuit **130**) selects the AC signal for ultrasonic transmission and causes the proximity detection unit **110** to perform ultrasonic transmission and reception, and when a distance between the object and the proximity detection unit **110** calculated by the calculation unit **152** becomes equal to or shorter than the predetermined distance, the signal applying unit selects the AC signal for capacitance detection and causes the proximity detection unit **110** to perform the capacitance detection, a simple configuration is realized in which one proximity detection unit **110**, one MUX **120**, one transmitter circuit **130**, and one receiver circuit **140** may be used for both the capacitance detection and the ultrasonic detection.

(76) The timer **145** is further provided to measure a period of time (round-trip time) from a time when the transmitter circuit **130** applies the second frequency signal to a time when the receiver circuit **140** measures electric charge based on the AC signal for ultrasonic transmission reflected by the object, and the calculation unit **152** calculates the distance between the object and the proximity

detection unit **110** based on the period of time measured by the timer **145**, and therefore, the distance to the object, that is, the hand, may be easily detected based on the round-trip times of the ultrasonic waves.

(77) Furthermore, the transmitter circuit **130** and the receiver circuit **140** may also perform switching between the AC signal for capacitance detection and the AC signal for ultrasonic wave detection in a time-division manner, and therefore, a simple configuration may be realized in which one transmitter circuit **130** and one receiver circuit **140** may be used for both the capacitance detection and the ultrasonic wave detection in a time-division manner.

(78) Since the proximity detection unit **110** at least includes one piezoelectric body **113** and one first electrode **111** and one second electrode **112** which are disposed in contact with the piezoelectric body **113**, as described above, it is not necessarily the case that a plurality of first electrodes **111** and a plurality of second electrodes **112** are required as illustrated in FIG. **1**. However, since distance data at a plurality of points is required to detect a two-dimensional profile or a three-dimensional image of the object, the proximity detection unit **110** may include a plurality of first electrodes **111** and a plurality of second electrodes **112** as illustrated in FIG. **1**. In this case, since the plurality of first electrodes **111**, one or more piezoelectric bodies **113**, and the plurality of second electrodes **112** are disposed and each of the one or more piezoelectric bodies **113** is disposed between at least one of the plurality of first electrodes **111** and at least one of the plurality of second electrodes **112**, the distance to the object, that is, the hand, may be measured by transmitting and receiving ultrasonic waves with the one or more piezoelectric bodies **113** disposed between the plurality of first electrodes **111** and the plurality of second electrodes **112**. Here, the number of piezoelectric bodies **113** is one when a sheet layer of the piezoelectric body **113** is disposed over an entire surface between a layer with the first electrodes **111** and a layer with the second electrodes **112**, for example.

(79) The first electrodes **111** extend in the X direction and are arranged in the Y direction that intersects with the X direction, the second electrodes **112** extend in the Y direction and are arranged in the X direction, and the piezoelectric body **113** is sandwiched between the first and second electrodes **111** and **112** at the intersections **110**A where the first and second electrodes **111** and **112** intersect. Accordingly, a configuration that may easily apply an AC signal for ultrasonic wave transmission to the piezoelectric body **113** and easily detect reflected waves may be realized using the first and second electrodes **111** and **112** for capacitance detection. Note that the capacitance detection may be performed by self-capacitance detection or by mutual capacitance detection. The first and second electrodes **111** and **112** may not intersect with each other, and a structure constituted such that a piezoelectric body **113** is sandwiched between first and second electrodes **111** and **112** may be arranged in a number of portions in a plane.

(80) Since the calculation unit **152** detects a two-dimensional profile or a three-dimensional image of the object based on the measured amount of electric charge, the proximity detection device **100** capable of easily recognizing a shape and a movement of the object, that is, the hand, near the operation surface **100**A may be provided. Specifically, the proximity detection device **100** may perform detailed image detection on an object by combining the capacitance detection and the ultrasonic detection, and may enable object detection at a wide range of distance by performing the capacitance detection in a region close to a touch and the ultrasonic detection in a region some distance away.

(81) Note that, although the proximity detection unit **110** includes the plurality of first electrodes **111**, the plurality of second electrodes **112**, and one or more piezoelectric bodies **113** in the foregoing description, the proximity detection unit **110** may be minimally configured with one first electrode **111**, one second electrode **112**, and one piezoelectric body **113**. The number of first electrodes **111** is not required to be equal to the number of second electrodes **112**.

(82) Furthermore, although the piezoelectric bodies **113** are disposed at the individual intersections **110**A of the plurality of first electrodes **111** and the plurality of second electrodes **112** in the

foregoing description, the piezoelectric bodies **113** may be disposed at only a number of the intersections **110**A. For example, the piezoelectric bodies **113** may be disposed at every other intersections **110**A in the X and/or Y direction. The number of piezoelectric bodies **113** is set according to application of the proximity detection device **100**, as the number relates to detection of a hand position using ultrasonic detection and resolution of the two-dimensional profile and the three-dimensional image.
(83) That is, each of the one or more piezoelectric bodies **113** is disposed between at least one of the plurality of first electrodes **111** and at least one of the plurality of second electrodes **112**.
(84) <Modification of Intersections **110**A>
(85) FIGS. **4**A to **7**C are diagrams illustrating modifications of the intersections **110**A. In FIGS. **4**A to **7**C, configurations of cross-sections corresponding to the cross-section of one of the intersections **110**A illustrated in FIG. **2** (the cross-section viewed from the II-II arrow in FIG. **1**) are illustrated. The intersection **110**A illustrated in FIG. **2** may be modified into the configuration shown in FIG. **4**A or FIG. **7**C.
(86) An intersection **110**A in FIG. **4**A has a first electrode **111**, a second electrode **112**, a piezoelectric body **113**, and a substrate **114**. The intersection **110**A in FIG. **4**A is configured without the uppermost substrate **114** of the intersection **110**A illustrated in FIG. **2**. For example, the piezoelectric body **113** and the first electrode **111** are disposed on the substrate **114** with the second electrode **112** on one surface thereof.
(87) An intersection **110**A in FIG. **4**B has a first electrode **111**, a second electrode **112**, a piezoelectric body **113**, and two substrates **114**, and is configured such that the piezoelectric body **113** at the intersection **110**A illustrated in FIG. **2** is made thinner.
(88) An intersection **110**A in FIG. **4**C has a first electrode **111**, a second electrode **112**, a piezoelectric body **113**, and two substrates **114**, and is configured such that the first and second electrodes **111** and **112** at the intersection **110**A illustrated in FIG. **2** are made thinner.
(89) An intersection **110**A in FIG. **4**D has a first electrode **111**, a second electrode **112**, piezoelectric bodies **113**, and two substrates **114**, and is configured such that the piezoelectric body **113** at the intersection **110**A illustrated in FIG. **2** is divided into two pieces.
(90) An intersection **110**A in FIG. **5**A has first electrodes **111**, a second electrode **112**, a piezoelectric body **113**, and two substrates **114**. In the cross-section illustrated in FIG. **5**A, the first electrode **111** of the intersection **110**A illustrated in FIG. **2** is divided into two pieces. The first electrode **111** may be divided into two pieces or the first electrode **111** may be shaped like a spiral in plan view, for example.
(91) An intersection **110**A in FIG. **5**B has first electrodes **111**, a second electrode **112**, piezoelectric bodies **113**, and two substrates **114**, and the piezoelectric body **113** illustrated in FIG. **5**A is divided into two pieces.
(92) An intersection **110**A in FIG. **5**C has a first electrode **111**, a second electrode **112**, a piezoelectric body **113**, two substrates **114**, and a shield electrode **115**. The shield electrode **115** is an example of a third electrode for shielding. In the intersection **110**A illustrated in FIG. **5**C, the second electrode **112** is disposed on the piezoelectric body **113** together with the first electrode **111**, and the shield electrode **115** is formed on the lower substrate **114**, and the piezoelectric body **113** is disposed on the shield electrode **115**. The shield electrode **115** is disposed on an opposite side of a side with the operation surface **100**A where an object, that is, a hand, approaches, relative to the first electrode **111** and the second electrode **112**.
(93) The first and second electrodes **111** and **112** are diamond-shaped patterned electrodes in plan view, and in FIG. **5**C, a bridge portion in which the first and second electrodes **111** and **112** cross over each other is omitted.
(94) The shield electrode **115** is disposed to shield the first and second electrodes **111** and **112** on the operation surface **100**A side from noise and to suppress parasitic capacitance generated relative to the ground, and an AC voltage may be applied to the shield electrode **115** or the shield electrode

115 may be connected to the ground. When an AC voltage is applied, the signal applying unit applies a third frequency signal to a third electrode (shield electrode **115**). The signal applying unit may also cause the shield electrode **115** to have a function of an active shield by setting a frequency of the third frequency signal to the same frequency as that of the first frequency signal when capacitance detection is performed. The shield electrode **115**, as an example, is formed of a metallic foil made of copper or aluminum or a conductive film made of a transparent conductive material, such as an ITO film. The shield electrode **115** is a single electrode disposed throughout the proximity detection unit **110** in plan view. An AC voltage is applied to the shield electrode **115** when a hand position is detected by capacitance.
(95) An intersection **110**A illustrated in FIG. **5**D is configured such that the piezoelectric body **113** of the intersection **110**A illustrated in FIG. **5**C is divided into two pieces so as to correspond to a first electrode **111** and a second electrode **112**. For example, ultrasonic waves may be transmitted by the first electrode **111** and received by the second electrode **112**.
(96) An intersection **110**A illustrated in FIG. **6**A is configured such that the piezoelectric body **113** under the second electrode **112** is removed from the intersection **110**A of FIG. **5**D. An intersection **110**A illustrated in FIG. **6**B is configured such that first and second electrodes **111** and **112** are thinner than those of the intersection **110**A of FIG. **6**A.
(97) An intersection **110**A illustrated in FIG. **7**A includes a first electrode **111**, a second electrode **112**, a piezoelectric body **113**, three substrates **114**, a shield electrode **115**, and an OCA (optical clear adhesive) **116**. The intersection **110**A illustrated in FIG. **7**A is configured such that, on the lowest one of the substrates **114**, the shield electrode **115**, the OCA **116**, one of the other substrates **114**, the second electrode **112**, the piezoelectric body **113**, the first electrode **111**, and the remaining substrate **114** are layered. In other words, the intersection **110**A of FIG. **7**A is configured such that the third substrate **114** with the shield electrode **115** disposed on one surface thereof is bonded with the OCA **116** under the substrate **114** on a lower side of the intersection **110**A illustrated in FIG. **2**. As with the intersection **110**A of FIG. **5**C, an AC voltage is applied to the shield electrode **115** when detection of a position of a hand is performed using capacitance.
(98) An intersection **110**A illustrated in FIG. **7**B is configured such that the uppermost substrate **114** of the intersection **110**A of FIG. **7**A is removed. In other words, the intersection **110**A of FIG. **7**B is configured such that the third substrate **114** with the shield electrode **115** disposed on one surface thereof is bonded with the OCA **116** under the intersection **110**A of FIG. **4**A.
(99) An intersection **110**A illustrated in FIG. **7**C is configured such that the substrate **114** under the second electrode **112** of the intersection **110**A of FIG. **7**A is removed and the substrate **114** with the shield electrode **115** disposed thereon is flipped upside down. The substrate **114** with the shield electrode **115** disposed thereon may be flipped upside down from that in FIG. **7**A, with the shield electrode **115** down, and bonded under the second electrode **112** with the OCA **116**.
(100) <Proximity Detection Device **100**M of Modification of Embodiment>
(101) FIG. **8** is a diagram illustrating an example of a proximity detection device **100**M according to a modification of the embodiment. The proximity detection device **100**M includes a proximity detection unit **110**M instead of the proximity detection unit **110** illustrated in FIG. **1**. The other configuration is the same as the proximity detection device **100** illustrated in FIG. **1**. Here, differences will be described.
(102) Instead of the configuration in which the piezoelectric bodies **113** are disposed at all the intersections **110**A between the first and second electrodes **111** and **112** as illustrated in the proximity detection unit **110** of FIG. **1**, the proximity detection unit **110**M is configured such that piezoelectric bodies **113**A for transmitting ultrasonic waves and piezoelectric bodies **113**B for receiving ultrasonic waves are disposed between the first and second electrodes **111** and **112** at intersections **110**B**1** and **110**B**2**, respectively.
(103) The different intersections **110**B**1** and **110**B**2** are arranged to include different first electrodes **111** in the plurality of first electrodes **111** every other first electrode **111**. Furthermore, the different

intersections **110**B**1** and **110**B**2** are arranged to include different first electrodes **111** in the plurality of second electrodes **112** every other second electrode **112**.
(104) The intersections **110**B**1** and **110**B**2** are not adjacent to each other in the X and Y directions in a plan view and are diagonally positioned. The piezoelectric bodies **113**A used for transmitting ultrasonic waves and the piezoelectric bodies **113**B used for receiving ultrasonic waves have the same configuration as the piezoelectric bodies **113** of the proximity detection device **100**.
(105) In the proximity detection device **100**M having this configuration, when ultrasonic waves are transmitted, the first and second electrodes **111** and **112** included in the intersections **110**B**1** are selected by the MUX **120** to apply an AC signal for ultrasonic waves to the first and second electrodes **111** and **112**, thereby applying the AC signal for ultrasonic waves to the piezoelectric bodies **113**A.
(106) Furthermore, when ultrasonic waves are received, the first and second electrodes **111** and **112** included in the intersections **110**B**2** are selected by the MUX **120**, and a waveform produced by charge between the first and second electrodes **111** and **112** is acquired by the receiver circuit **140**.
(107) A method for performing position detection using capacitance in the proximity detection device **100**M is the same as the method for performing position detection using capacitance in the proximity detection device **100**.
(108) By separating the piezoelectric bodies **113**A used for transmitting ultrasonic waves from the piezoelectric bodies **113**B used for receiving ultrasonic waves, as in the proximity detection device **100**M according to the modification of the embodiment, the switching control performed by the MUX **120** and control of the waveform acquisition performed by the receiver circuit **140** are simplified when a distance to a hand is detected using ultrasonic waves. In addition, since the functions of transmitter and receiver are separated from each other, it is advantageous that performance of the device is easily improved.
(109) Although the exemplary embodiment of the proximity detection device according to the present invention has been described hereinabove, the present invention is not limited to the embodiment disclosed in detail, and various modifications and changes may be made without departing from claims.

Claims

1. A proximity detection device, comprising: a proximity detection unit capable of detecting proximity of an object, based on a capacitance formed between the object and the proximity detection unit and an ultrasonic wave transmitted to and reflected from the object, the proximity detection unit including: at least one piezoelectric body; and at least one first electrode and at least one second electrode disposed in contact with the at least one piezoelectric body, such that the piezoelectric body is disposed between the first electrode and the second electrode at an intersection of the first electrode and second electrode, while the first electrode and the second electrode are insulated from each other without contacting with the piezoelectric body at portions other than the intersection; a signal applying unit configured to cause the proximity detection unit to perform capacitance detection, or to perform ultrasonic detection by transmitting and/or receiving ultrasonic waves, by selectively applying a plurality of signals of different frequencies to at least one of the first and second electrodes, the plurality of signals including a first frequency signal for the capacitance detection and a second frequency signal for the ultrasonic detection; a charge measurement unit connected to at least one of the first and second electrodes, configured to measure an electric charge on the at least one of the first and second electrodes; and a calculation unit configured to calculate a distance between the object and the proximity detection unit, wherein the signal applying unit is further configured, when the second frequency signal has been applied to cause the proximity detection unit to perform the ultrasonic detection, and if the distance between the object and the proximity detection unit calculated by the calculation unit is equal to or smaller

than a predetermined distance, to select the first frequency signal instead and cause the proximity detection unit to perform the capacitance detection.

2. The proximity detection device according to claim 1, wherein the proximity detection unit is configured to transmit an ultrasonic wave and then receive a reflected ultrasonic wave which is the ultrasonic wave reflected by the object.

3. The proximity detection device according to claim 1, wherein the calculation unit is further configured to obtain a result of the capacitance detection and/or a result of the ultrasonic detection, based on the first or second frequency signal selected by the signal applying unit and the electric charge measured by the charge measurement unit, and calculates the distance between the object and the proximity detection unit based on the obtained result.

4. The proximity detection device according to claim 3, wherein the calculation unit obtains the result of the capacitance detection which is an amount of the electric charge measured by the charge measurement unit, when the first frequency signal is selected by the signal applying unit.

5. The proximity detection device according to claim 3, wherein the calculation unit obtains the result of the ultrasonic detection which is a period of time between the transmitting an ultrasonic wave and receiving a reflected ultrasonic wave which is the ultrasonic wave reflected by the object, when the second frequency signal is selected by the signal applying unit.

6. The proximity detection device according to claim 3, wherein the signal applying unit is further configured to switch between applying the first frequency signal and applying the second frequency signal in a time-division manner.

7. The proximity detection device according to claim 3, further comprising: a timer configured to measure a period of time from an application of the second frequency signal by the signal applying unit to a measurement of the electric charge based on reception of an ultrasonic wave which corresponds to the second frequency signal and has been reflected by the object, wherein the calculation unit is further configured to calculate the distance between the object and the proximity detection unit based on the period of time measured by the timer.

8. The proximity detection device according to claim 3, wherein the calculation unit is configured to detect a two-dimensional profile or a three-dimensional image of the object, based on an amount of electric charge measured by the charge measurement unit and/or a period of time from transmitting the ultrasonic wave to receiving the ultrasonic wave reflected by the object.

9. The proximity detection device according to claim 1, wherein the proximity detection unit comprises: a plurality of first electrodes; and a plurality of second electrodes.

10. The proximity detection device according to claim 9, wherein each of the plurality of first electrodes extends in a first direction, the plurality of first electrodes being arranged side by side in a second direction that intersects with the first direction, wherein each of the plurality of second electrodes extends in the second direction, the plurality of second electrodes being arranged side by side in the first direction, and wherein a plurality of piezoelectric bodies are sandwiched between the plurality of first and second electrodes at positions where the first and second electrodes intersect with each other.

11. The proximity detection device according to claim 1, further comprising: a third electrode disposed on an opposite side of the first and second electrodes opposite to a detection side of the first and second electrodes which the object approaches.

12. The proximity detection device according to claim 11, wherein the third electrode is connected to a ground.

13. The proximity detection device according to claim 11, wherein the signal applying unit is further configured to apply a third frequency signal to the third electrode.

14. The proximity detection device according to claim 1, wherein the piezoelectric body is an electret having piezoelectric properties.